String Theory and M-Theory

Lecture 9: Strings, Gravity, Compactification, and T-Duality

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Companion notes curated by LazyingArt LLC with Video2Book

Chapter 1

Strings, Gravity, Compactification, and T-Duality

Overview. A point particle can move consistently on essentially any smooth geometry, but a quantum string cannot. Its infinitely many zero-point oscillations probe the target space at successively shorter scales, and the effective geometry changes as those oscillations are restored. Requiring that this process have a fixed point gives $R_{\mu\nu} = 0$, the vacuum Einstein equation. We then turn from this consistency condition to compactification. A circle supplies both Kaluza–Klein momentum states and string winding states; exchanging them identifies a circle of radius R with one of radius α'/R.

1.1 Consistency as a Selection Principle

There are two questions to answer. Why does gravity occur in string theory, and why is string theory far less permissive than point-particle mechanics? They are really one question. A background geometry is acceptable only if a string can propagate on it without retaining an arbitrary dependence on how finely we resolve the string.

Begin with the control problem: a point particle constrained to a smooth curved surface. Classically it follows a geodesic, the intrinsic analogue of a straight line. One can approximate the motion by many short segments, write a difference equation, and then let the segment length tend to zero. The result is an ordinary differential equation with a well-defined continuum limit.

Quantum mechanics changes the sum over histories, not the conclusion. Instead of selecting one classical path, we sum amplitudes over paths on the surface. Time may again be divided into small steps, and the limit is equivalently described by a Schrödinger equation on the curved configuration space. For a smooth surface, the limit settles to a definite propagator and spectrum.

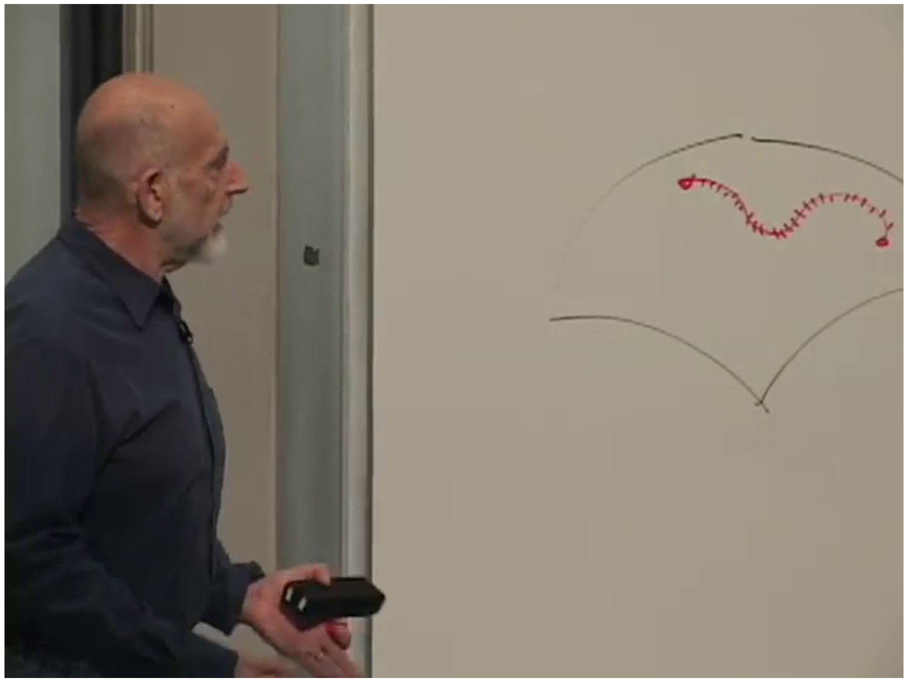

Figure 1.1: A smooth path on the surface and a jagged path from the time-sliced quantum sum. Both remain intrinsic to the target surface.

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Question from the audience. In the path integral, are all of the paths required to remain inside the surface? [a]

Response. Yes. The surface is the particle's configuration space, not a sheet embedded in a larger room through which the particle may take a shortcut. Every intermediate point belongs to the surface, and all lengths and actions are computed with its intrinsic metric.

[a] Lecture timestamp: 00:05:39.

The corresponding question for a string is harder. A string is not specified only by a center-of-mass position. It carries an infinite tower of oscillators, so improving the resolution means admitting more dynamical degrees of freedom. The theory is consistent only if observables approach a limit as that tower is restored.

1.2 The Point Particle on a Sphere

The sphere is the simplest curved test geometry. Let its radius be R, and first place a particle of mass m on it. For motion along a great circle,

$$E = \frac{1}{2}mv^2 = \frac{p^2}{2m}, \qquad L = pR. \tag{1.1}$$

Consequently,

$$E = \frac{L^2}{2mR^2} = \frac{L^2}{2I}, \qquad I = mR^2. \tag{1.2}$$

The radius enters through the moment of inertia. Quantizing the angular momentum produces the familiar rigid-rotor spectrum, and nothing depends on an ultraviolet cutoff.

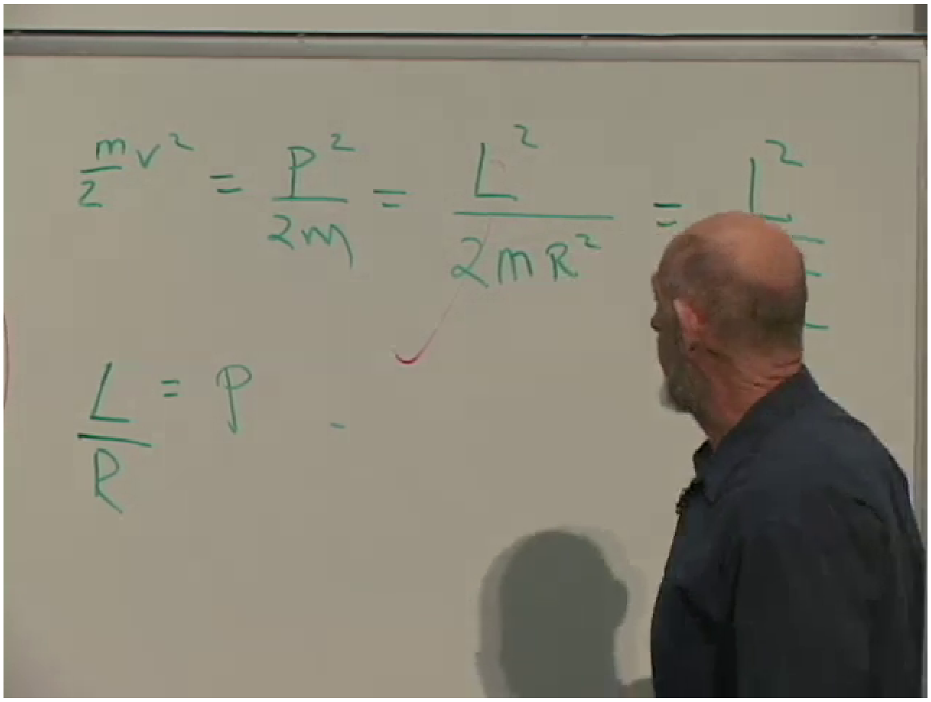

Figure 1.2: The point-particle control calculation: $p^2/(2m) = L^2/(2mR^2) = L^2/(2I)$.
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This untroubled example is important. Curvature by itself is not the problem. The problem appears when curvature is coupled to the internal quantum extension of a string.

1.3 The Vacuum String Is Not Pointlike

Consider one transverse coordinate of an open string and subtract its center of mass. The oscillator expansion used here is schematic:

$$X(\sigma) - X_{\rm cm} \sim \sum_{n=1}^{\infty} \frac{a_n + a_n^\dagger}{\sqrt{n}} \cos(n\sigma). \tag{1.3}$$

The essential ingredients are the cosine modes, the oscillator operators, and the factor $1/\sqrt{n}$. Exact normalizations play no role in the size argument. [1]

[1] Editorial clarification: The precise expansion contains convention-dependent factors of α', numerical constants, and a choice of oscillator notation. Equation (1.3) retains only the dependence needed for the ultraviolet estimate, just as in the board derivation.

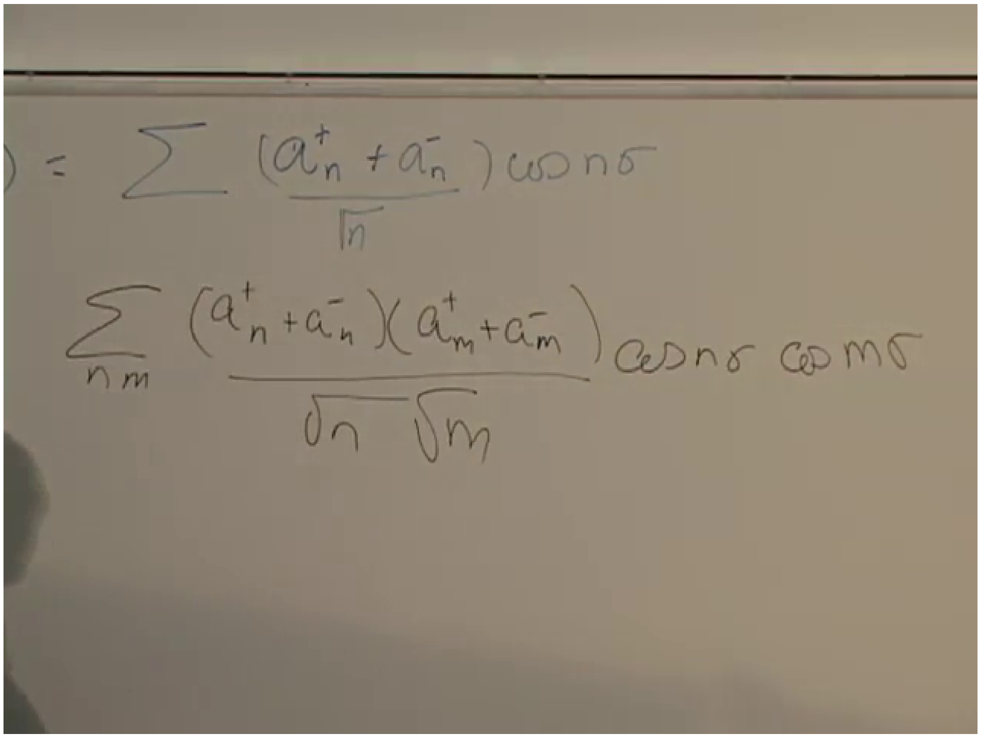

Figure 1.3: The structural oscillator expansion and the double sum obtained on squaring it.
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The squared distance from the center of mass in the oscillator vacuum is

$$\begin{aligned} \left\langle (X(\sigma) - X_{\text{cm}})^2 \right\rangle_0 \sim \sum_{n,m\geq 1} & \frac{\cos(n\sigma)\cos(m\sigma)}{\sqrt{nm}} \\ & \times \langle 0| \, (a_n + a_n^\dagger) \\ & \qquad \times (a_m + a_m^\dagger) \, |0\rangle \, . \end{aligned} \tag{1.4}$$

Most terms vanish. Two annihilation operators kill the ket; two creation operators leave a state orthogonal to the bra; and a creation operator followed by an annihilation operator also has zero vacuum expectation in this ordering. The surviving contraction is

$$\langle 0| \, a_n a_m^\dagger \, |0\rangle = \delta_{nm}. \tag{1.5}$$

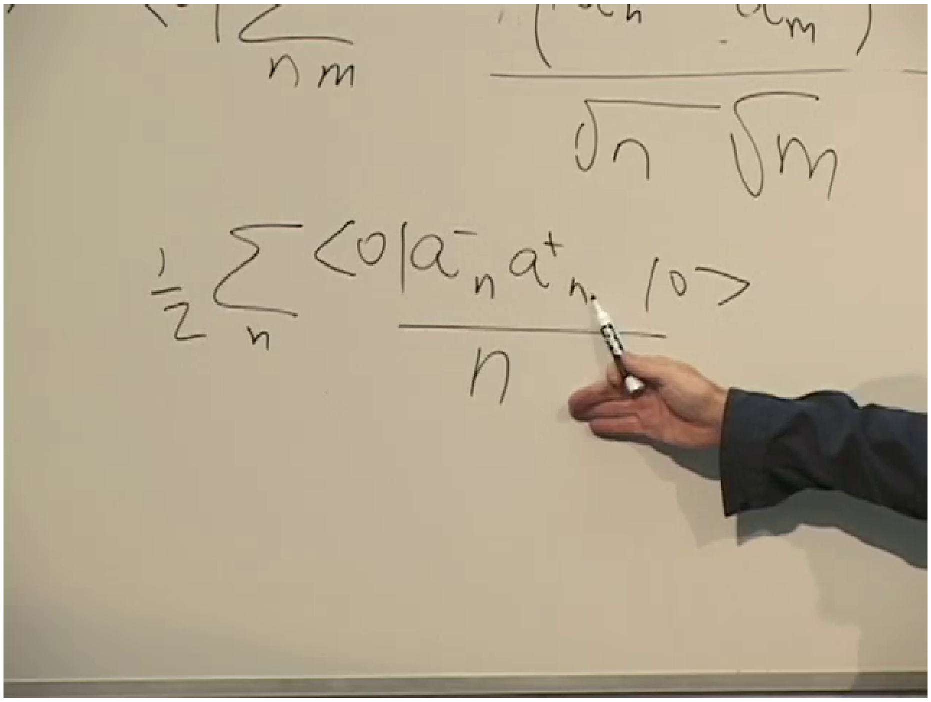

Figure 1.4: The surviving vacuum contraction collapses the double oscillator sum to a single sum.
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Thus

$$\left\langle (X(\sigma) - X_{\rm cm})^2 \right\rangle_0 \sim \sum_{n=1}^{\infty} \frac{\cos^2(n\sigma)}{n}. \tag{1.6}$$

At a generic point along the string, or after averaging over σ, $\cos^2(n\sigma)$ contributes roughly 1/2. With a mode cutoff N,

$$\left\langle (X - X_{\rm cm})^2 \right\rangle_{0,N} \sim \frac{1}{2}\sum_{n=1}^{N} \frac{1}{n} = \frac{1}{2}\log N + O(1). \tag{1.7}$$

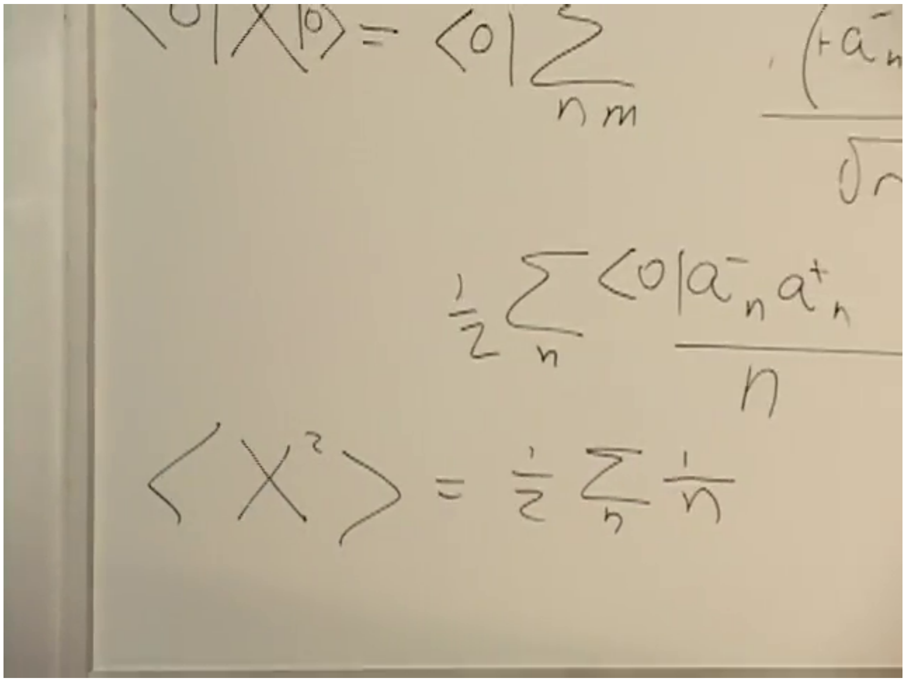

Figure 1.5: The board arrives at the harmonic sum controlling the ground-state spread.
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Keeping only, say, the first twenty-five modes gives a finite fuzzy object, but it is not yet the continuum string. Removing the cutoff causes the mean-square spread to grow logarithmically. This conclusion should not be misread as a proof that flat-space string theory is inconsistent. In flat space the center-of-mass and oscillator motions separate, and delicate cancellations in physical amplitudes preserve the expected locality. The growing size is a warning. Curvature tells us whether the warning becomes a failure.

1.4 When Internal Spreading Changes the Geometry

Place the string near the north pole of the sphere. With every oscillator turned off, it behaves like the point particle just studied. As higher modes are admitted, the string becomes a fluctuating tangle spread over the cap. Its center may still travel around the sphere, but the object whose moment of inertia enters the rotational energy is now extended.

For an axis through the center of the sphere and the poles,

$$I = \int r_\perp^2 \, dm, \tag{1.8}$$

where $r_\perp$ is the perpendicular distance from a string element to that axis. A point at the equator of the orbit has the largest relevant lever arm. Portions of a string that droop toward the axis contribute less. At fixed total mass this lowers the effective moment of inertia:

$$I_{\text{eff}} < mR^2. \tag{1.9}$$

The rotational energy therefore looks like that of a point object on a smaller sphere,

$$E_{\text{rot}} = \frac{L^2}{2I_{\text{eff}}} \equiv \frac{L^2}{2mR_{\text{eff}}^2}, \qquad R_{\text{eff}} < R. \tag{1.10}$$

Question from the audience. Why should the moment of inertia tend toward zero if the string is spreading over a larger region? [a]

Response. The statement is about the effective description under successive coarse-graining, not a claim that the raw extended distribution immediately has zero inertia. The first band of modes turns the point into a fuzzy lump that moves as though it lived on a smaller sphere. The next band resolves structure inside each such lump and repeats the same renormalization. Iterating the comparison drives the effective radius inward.

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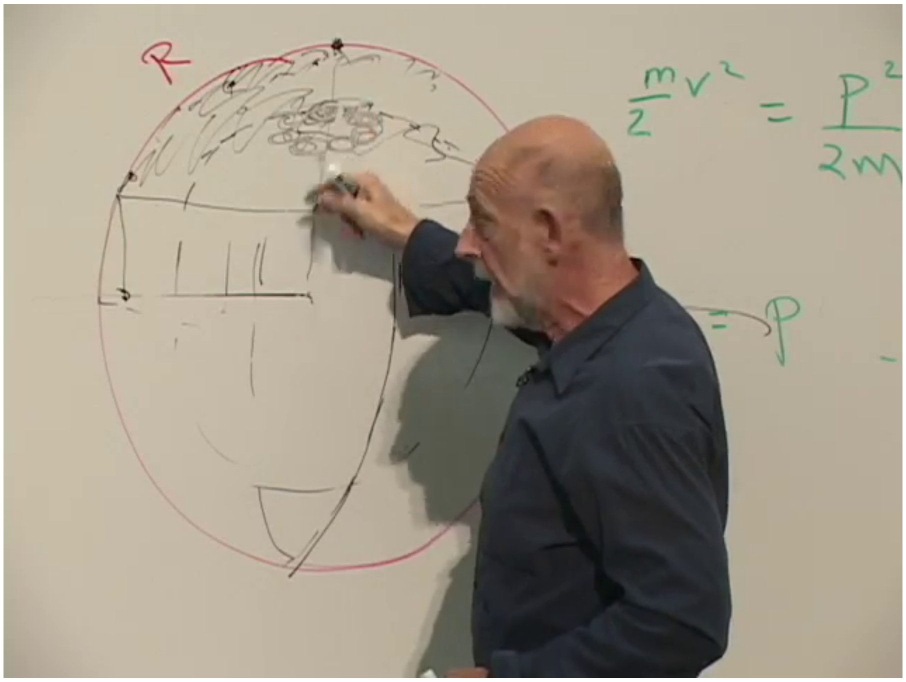

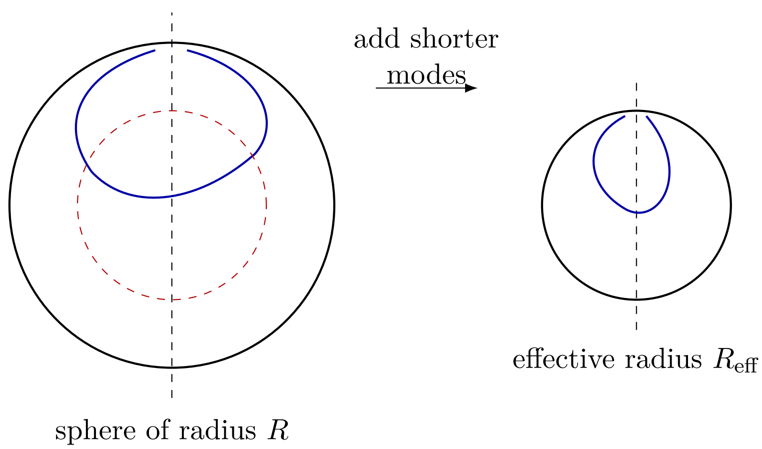

Figure 1.6: The north-pole string on the board and a clean reconstruction of the effective-radius argument. The dashed inner circle is an effective geometry, not a second physical shell inside the sphere.

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This is a renormalization-group statement. A finite cutoff makes the string a lump with finite internal resolution. Raising the cutoff changes the mechanics of that lump. On a positively curved sphere each added shell of modes makes the background inferred from the long-distance motion smaller. If the result never stabilizes, there is no cutoff-independent string theory on that background.

1.5 The Metric Must Reach a Fixed Point

Now replace the sphere by a general target-space metric $g_{\mu\nu}(x)$. Integrating in a thin logarithmic shell of oscillator modes changes that metric by a local symmetric rank-two tensor. The change must vanish in flat space and begin with two derivatives of the metric. At leading order, covariance leaves the Ricci tensor as the relevant candidate:

$$\frac{\partial g_{\mu\nu}}{\partial \log \Lambda} = -C\, R_{\mu\nu} + \cdots\,, \qquad C > 0 \tag{1.11}$$

in the direction of coarse-graining used in the geometric argument. Positive curvature then drives a sphere inward, as required. [2]

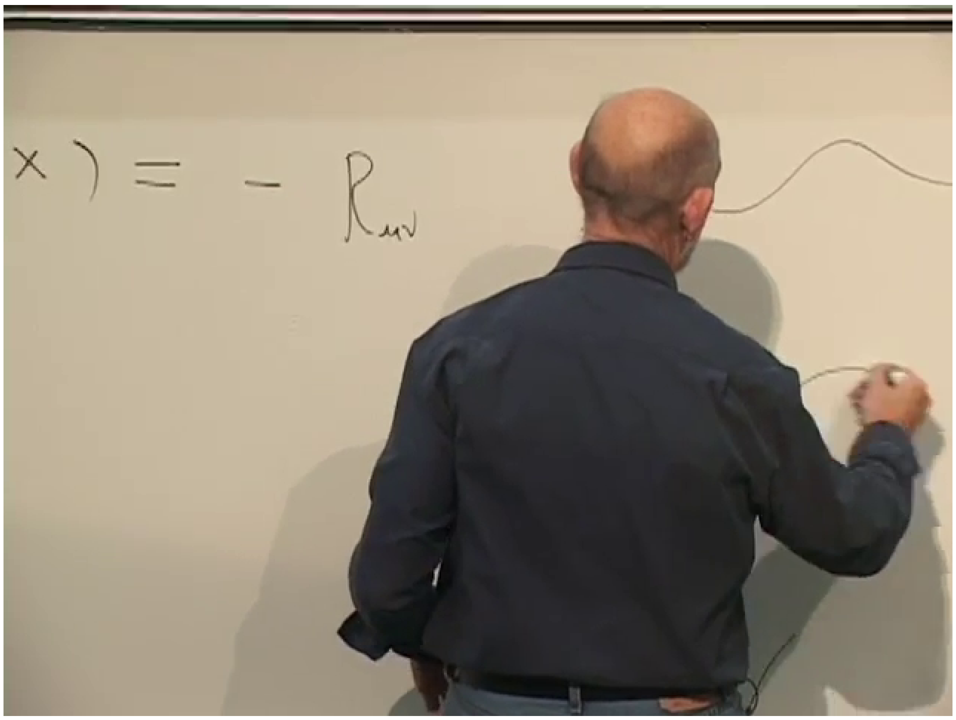

Figure 1.7: The metric change is written schematically as minus the Ricci tensor, with a smoothing curve drawn beside it.

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Question from the audience. Where does the oscillator cutoff N appear in this flow equation? [a]

Response. It is the scale variable. The string-size sum changes logarithmically when the highest included mode changes, so the natural evolution parameter is $\log N$, or equivalently $\log \Lambda$. Equation (1.11) describes the infinitesimal change caused by one logarithmic shell of modes.

[a] Lecture timestamp: 00:43:38.

An admissible background is a fixed point of this flow. To leading order,

$$\boxed{R_{\mu\nu} = 0.} \tag{1.12}$$

[2] Editorial clarification: The normalization and sign depend on whether the renormalization parameter increases toward the ultraviolet or toward the infrared. In conventional worldsheet notation the leading metric beta function is $\beta^{g}_{\mu\nu} = \alpha' R_{\mu\nu} + \cdots$. The invariant statement used here is that cutoff independence requires the beta function, and hence $R_{\mu\nu}$ at this order, to vanish.

Ricci flat does not mean that every component of the Riemann tensor vanishes. Gravitational waves, for example, can have nonzero curvature in vacuum while satisfying $R_{\mu\nu} = 0$.

1.5.1 The Same Condition Is the Vacuum Einstein Equation

With conventional constants suppressed, Einstein's equation is

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = T_{\mu\nu}. \tag{1.13}$$

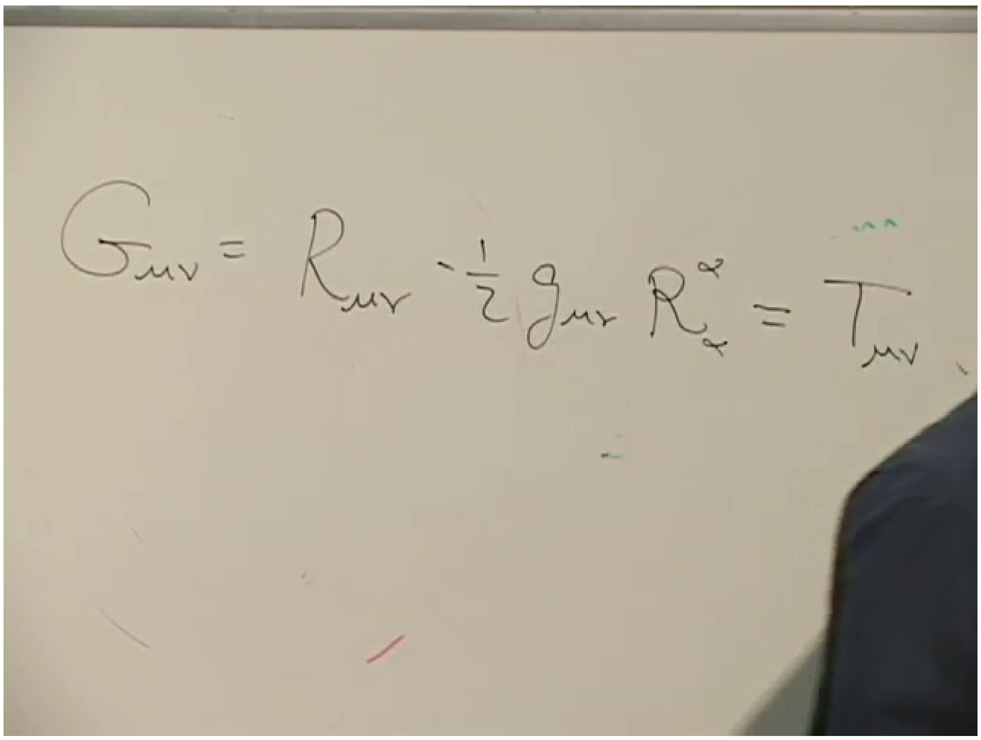

Figure 1.8: The Einstein tensor and stress tensor on the board.
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In vacuum $T_{\mu\nu} = 0$. Taking the trace in D spacetime dimensions gives

$$0 = g^{\mu\nu} G_{\mu\nu} = \left(1 - \frac{D}{2}\right) R. \tag{1.14}$$

For $D \neq 2$, $R = 0$, and substitution back into (1.13) yields $R_{\mu\nu} = 0$. Thus the condition that the string lose its cutoff dependence is the vacuum gravitational field equation. Gravity has not been appended to the theory; it has appeared as the price of consistency.

The word *vacuum* does not mean that nothing happens. It means there is no stress-energy source in the region. Source-free Maxwell equations admit electromagnetic waves, and source-free Einstein equations admit gravitational waves. The geometry may be dynamical even when its Ricci tensor vanishes.

Question from the audience. Is this condition connected with conformal invariance? [a]

Response. Yes, although the question is partly lost in the recording. The relevant symmetry is conformal invariance of the *worldsheet* quantum theory. Requiring the worldsheet scale anomaly to vanish is what sets the metric beta function to zero. It should not be confused with demanding conformal invariance of spacetime itself.

[a]Lecture timestamp: 00:52:20.

There is a second consistency condition. Worldsheet conformal invariance also fixes the critical spacetime dimension: ten for superstrings and twenty-six for the bosonic string. Ricci flatness alone is therefore not the complete list of requirements, but it is the part that displays the Einstein equation most directly.

1.6 Compact Dimensions Rather Than Missing Dimensions

A superstring background has nine spatial dimensions and one time dimension. Our macroscopic world displays only three large spatial directions, so six spatial dimensions must be hidden or otherwise rendered inaccessible at ordinary scales. Compactification keeps them as genuine directions while giving them small finite extent.

Question from the audience. If different observers carry different clocks, does that amount to having more than one time dimension? [a]

Response. No. Many clocks are many instruments measuring coordinates in one spacetime; they do not create independent timelike directions. The extra dimensions considered here are spatial. The familiar quantum and causal constructions do not supply a comparable consistent theory with several local time directions, so that possibility is not part of this compactification model.

[a]Lecture timestamp: 00:56:59.

Imagine creatures who believe their world is a line. At low resolution they can move only forward and backward. Under a sufficiently powerful microscope, however, the line is revealed to be a very thin cylinder. A small creature can move around its circular direction as well as along its length. The second direction was always present; its circumference was too small for coarse probes to resolve.

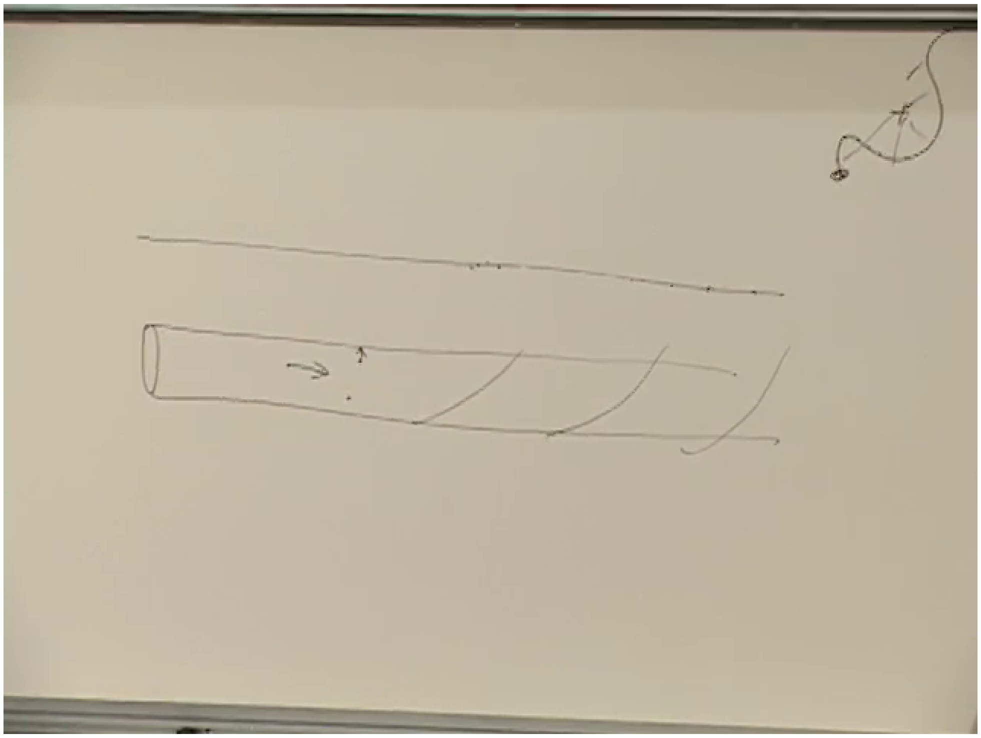

Figure 1.9: A narrow cylinder and trajectories that wind around its hidden circular direction.
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A cylinder can be represented as a strip whose upper and lower edges are identified. Crossing the upper edge returns through the lower edge at the same longitudinal position. There is no physical boundary at either edge. The same operation can be performed with two directions: identify the left and right edges of a rectangle and also its top and bottom edges. The result is a two-torus.

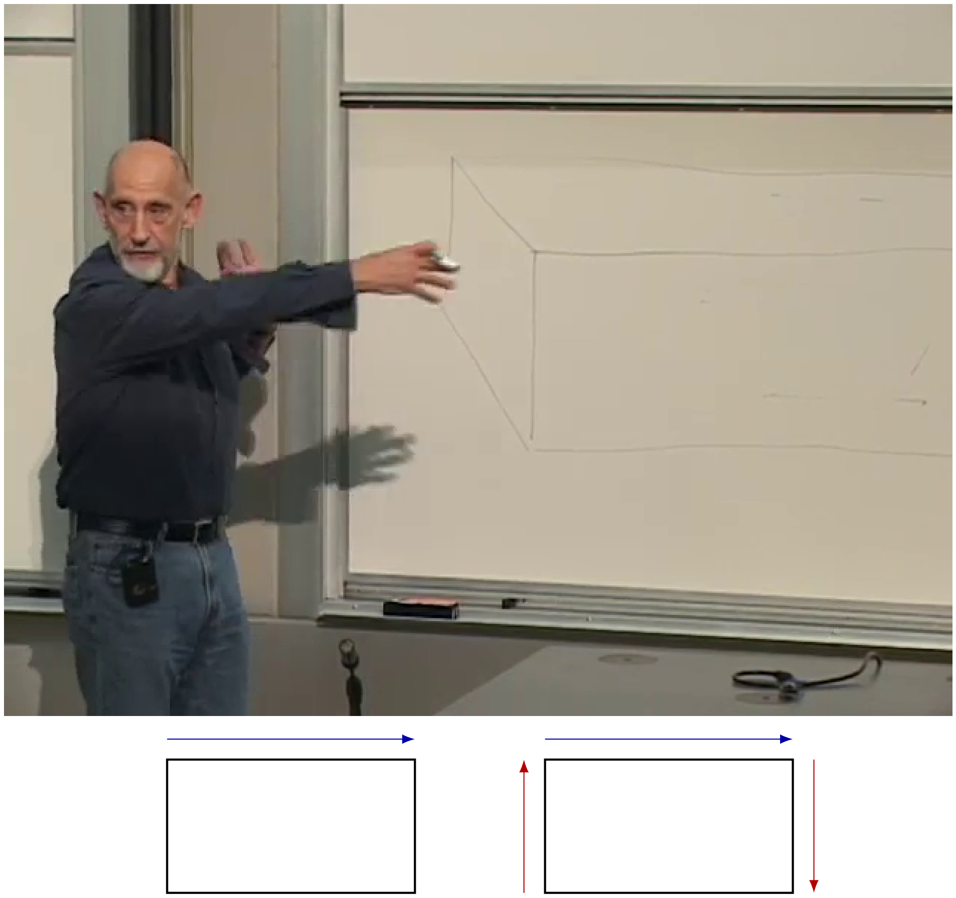

Figure 1.10: The board's slab and a clean fundamental-domain description. Identifying one edge pair gives a cylinder; identifying both pairs gives a torus.

Lecture frame: 01:04:40.

The torus is a topology before it is a bagel-shaped surface in three-dimensional space. Cut a torus once and it becomes a cylinder; cut again and it becomes a rectangle, provided that the edge identifications are remembered. A flat metric may be put on this rectangle, so the resulting flat torus has zero Riemann curvature locally and is certainly Ricci flat.

A straight trajectory on the rectangle continues straight across an identified edge. If its slope is a rational ratio of the two periods, the trajectory eventually closes. If the ratio is irrational, it never closes and becomes dense on the torus.

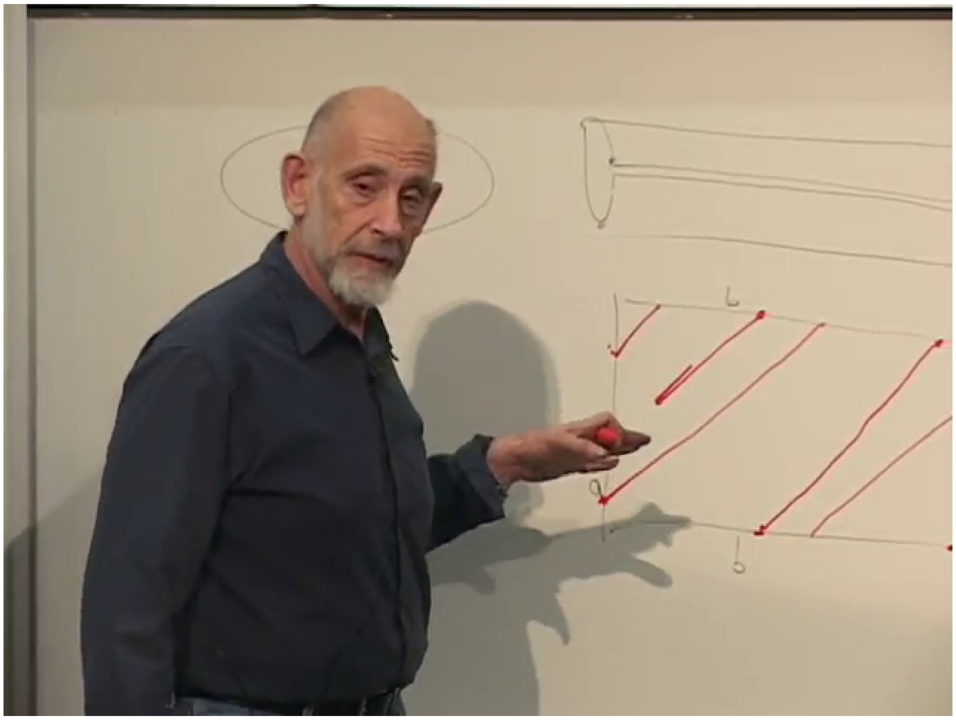

Figure 1.11: Repeated straight segments in a fundamental domain represent one uninterrupted trajectory on the torus.

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Higher-dimensional tori are formed in the same way. A cube with each pair of opposite faces identified is a three-torus, and six compact directions can be modeled by a six-torus. One should not try to embed all of these objects in ordinary three-dimensional space; the face-identification description is intrinsic and complete.

Question from the audience. Must an extra dimension necessarily come back on itself? [a]

Response. Not as a theorem. It could be noncompact, but then it would normally appear as another large direction in the observed long-distance world. Compactness is the simplest way to reconcile extra dimensions with their apparent absence. Why nature chooses a particular compactification is a separate dynamical question.

[a]Lecture timestamp: 01:15:03.

Question from the audience. Why assume the unseen dimensions are ordinary spatial directions curled up in this way rather than something altogether different? [a]

Response. Because this is a definite solvable hypothesis, not because it has been proved to be the unique description of nature. Compact spatial manifolds let us construct consistent examples and compute their spectra. More exotic possibilities must earn their place by doing the same.

[a]Lecture timestamp: 01:15:51.

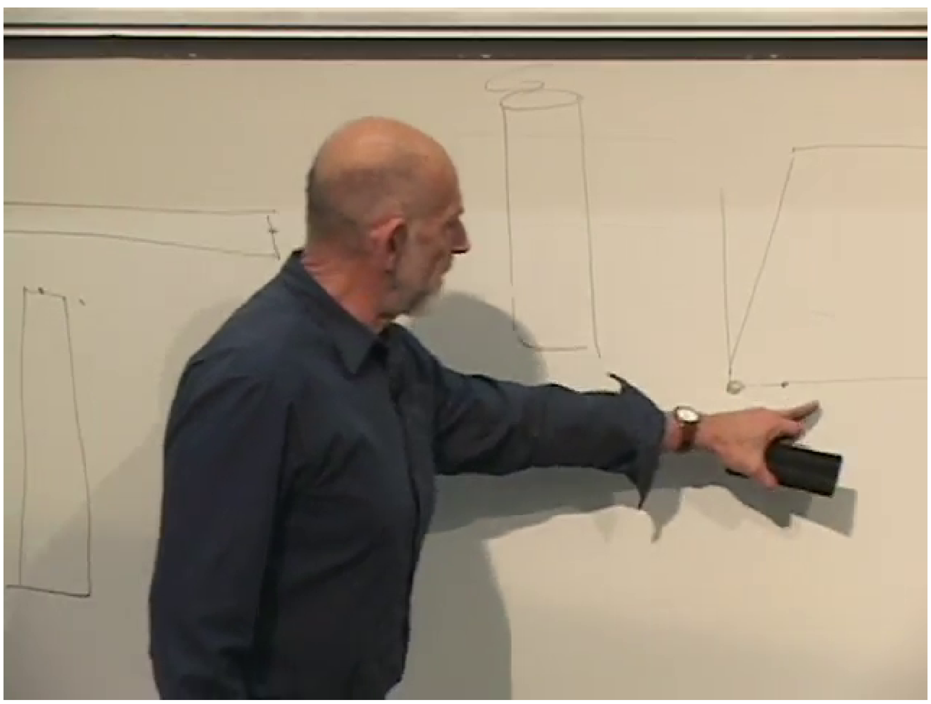
Question from the audience. Distances on the familiar bagel-shaped torus do not look like distances in a rectangle. Which geometry is meant? [a]

Response. The term *torus* fixes the identifications, not one unique metric. The embedded bagel inherits a curved metric from three-dimensional space. The periodically identified rectangle carries a flat metric. They have the same topology and different geometry; for the simple compactification we mean the flat one.

[a]Lecture timestamp: 01:18:11.

1.7 The Shape Parameters of a Flat Torus

Even a flat two-torus is not unique. Its fundamental parallelogram has an overall area, an aspect ratio, and a shear. Changing these parameters changes distances and therefore changes the spectrum, without changing the topology.



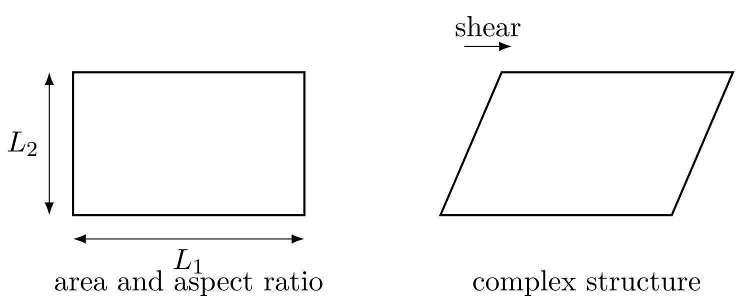


Figure 1.12: The rectangular and sheared fundamental domains used to discuss torus moduli.
Lecture frame: 01:23:00.

For a two-torus it is useful to package the shape into a complex number $\tau = \tau_1 + i\tau_2$: the real part records shear and the imaginary part records an aspect ratio after an overall scale is separated. Interchanging the two basic cycles or changing their integer basis can make apparently different parallelograms describe the same torus.

Question from the audience. What happens if one edge is flipped before the identification? [a]

Response. That produces a different topology. Identifying with a reflection gives a nonorientable surface, the Klein bottle, rather than an ordinary torus.

[a] Lecture timestamp: 01:24:20.

Question from the audience. What if the twist is not uniform along the edge? [a]

Response. A smooth nonuniform deformation can be transferred into the metric on the fundamental domain. The topological identification and the geometrical data must be kept distinct: changing the gluing may change topology, while smoothly changing distances changes the metric.

[a] Lecture timestamp: 01:25:01.

[3]

Flat tori are useful because they solve the consistency condition immediately, but their symmetry is too generous for a realistic model of particle physics. More intricate Ricci-flat spaces, notably Calabi–Yau manifolds, provide richer compactifications. Their detailed construction lies beyond this discussion; the principle remains the same.

1.8 Reading a Compact Circle from Its Spectrum

After the break, reduce the compact space to one circle of circumference $2\pi R$. A wavefunction must be single-valued after one circuit:

$$e^{ip_c(x+2\pi R)/\hbar} = e^{ip_c x/\hbar}. \tag{1.15}$$

Hence

$$p_c R = n\hbar, \qquad p_c = \frac{n\hbar}{R}, \qquad n \in \mathbb{Z}. \tag{1.16}$$

[3] Editorial clarification: For a flat two-torus, the area is the simplest Kahler modulus and τ is the complex-structure modulus. A continuous shear is part of τ. A Dehn twist is instead a discrete mapping-class operation, represented by $\tau \mapsto \tau + 1$; it is not itself the continuous modulus. This sharpens the intentionally informal terminology used during the board discussion.

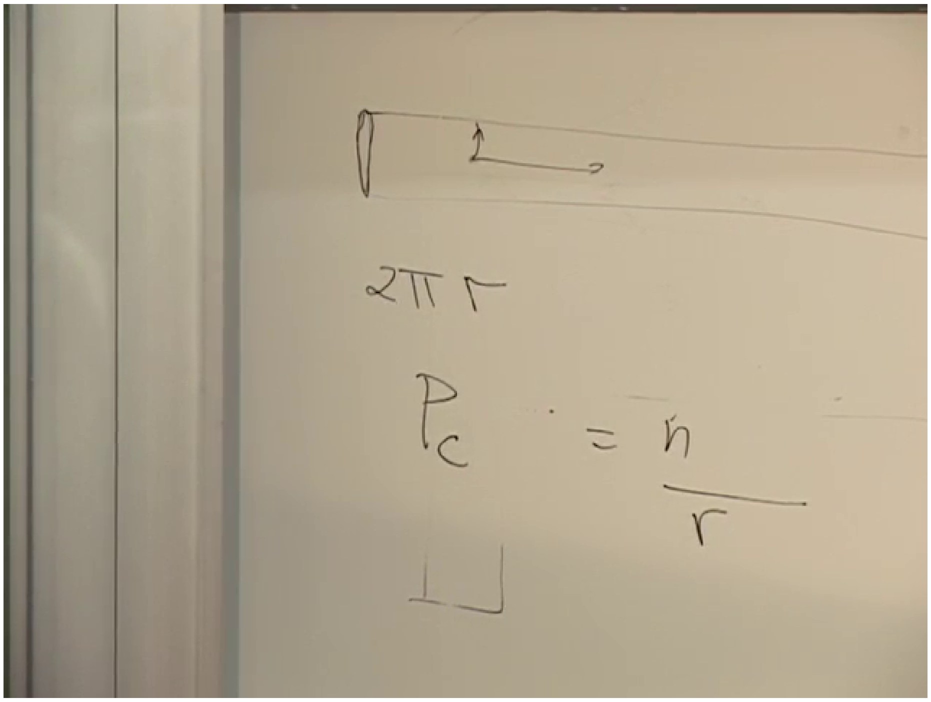


Figure 1.13: The compact circumference $2\pi R$ and quantized momentum $p_c = n\hbar/R$.
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For a massless particle in the full spacetime,

$$E = c|\mathbf{P}|. \tag{1.17}$$

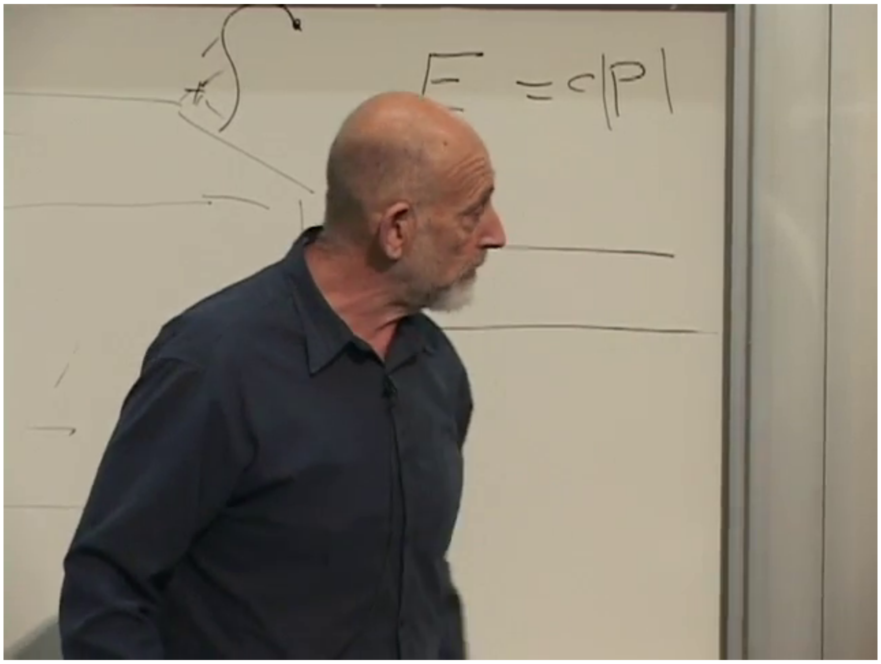

Figure 1.14: The massless dispersion relation used to reinterpret compact momentum as lower-dimensional mass.

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Split the momentum into noncompact and compact parts and set $c = \hbar = 1$:

$$E^2 = \mathbf{p}_{\text{large}}^2 + \frac{n^2}{R^2}. \tag{1.18}$$

An observer who cannot resolve the compact direction compares this with $E^2 = \mathbf{p}_{\text{large}}^2 + m^2$ and infers a tower of masses

$$m_n = \frac{|n|}{R}. \tag{1.19}$$

The two signs of n describe opposite compact momenta with the same positive energy, not positive and negative masses. The spacing $1/R$ therefore reveals the radius spectroscopically. This is the Kaluza–Klein mechanism: momentum in a hidden direction appears as mass in the visible dimensions.

1.9 Winding States Complete the Spectrum

A closed string has a possibility unavailable to a point particle: it can wrap around the circle. If the winding number is w, the string closes only after covering the compact direction w times. Its length is $2\pi|w|R$, so tension supplies an energy

$$m_w = 2\pi T\,|w|R = \frac{|w|R}{\alpha'}. \tag{1.20}$$

The lecture chooses units with $\alpha' = 1$, reducing this to

$$m_w = |w|R. \tag{1.21}$$

Positive and negative w are opposite orientations around the circle.

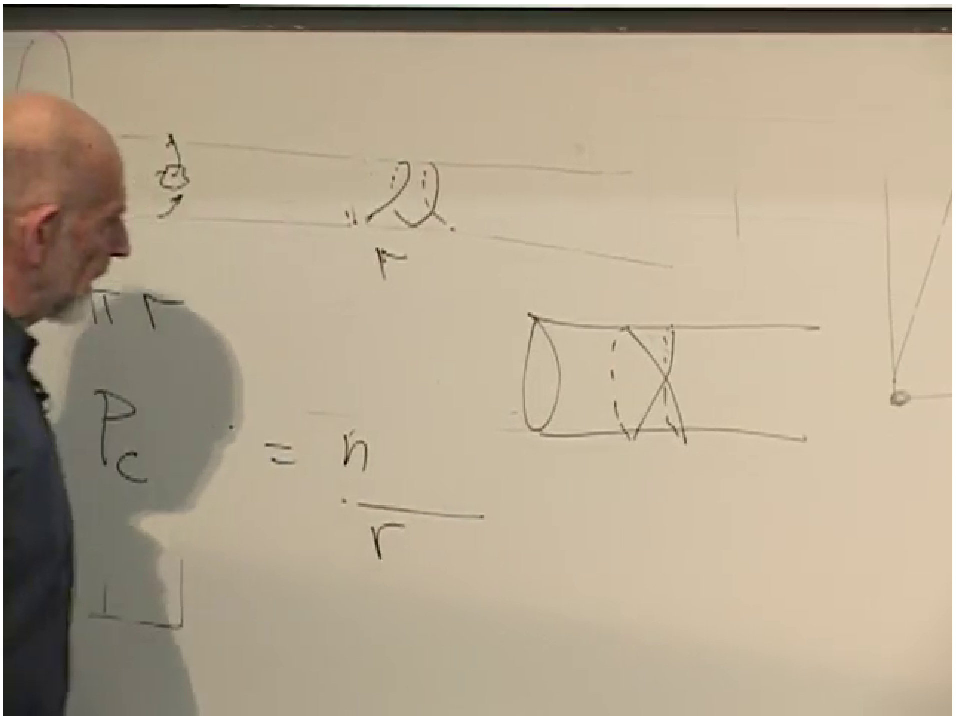

Figure 1.15: A closed string wound around the compact cylinder, with the compact radius marked on the board.

Lecture frame: 01:39:45.

The compact circle thus produces two complementary ladders:

$$m_n = \frac{|n|}{R}, \qquad m_w = \frac{|w|R}{\alpha'}. \tag{1.22}$$

When R is large, momentum levels are closely spaced and winding states are heavy. When R is small, momentum levels are widely spaced and winding states are light.

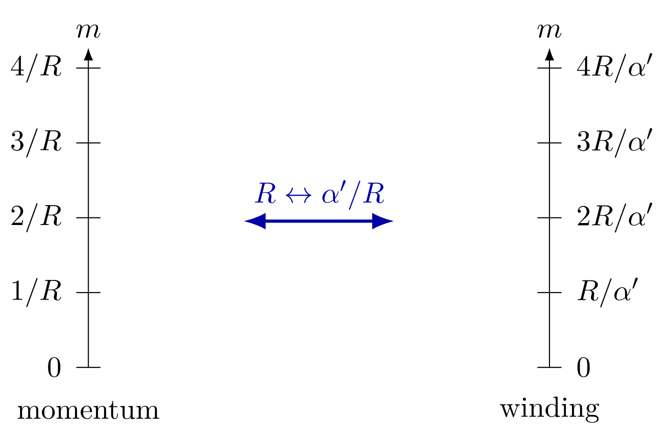

Figure 1.16: Momentum and winding towers exchange when the radius is inverted in string units.

1.10 T-Duality: Small and Large Circles Are the Same Physics

The two towers are interchanged by

$$R \longleftrightarrow \frac{\alpha'}{R}, \qquad n \longleftrightarrow w. \tag{1.23}$$

In the units used on the board, $\alpha' = 1$, so this is simply $R \leftrightarrow 1/R$. At the self-dual radius $R = \sqrt{\alpha'}$, momentum and winding scales coincide.

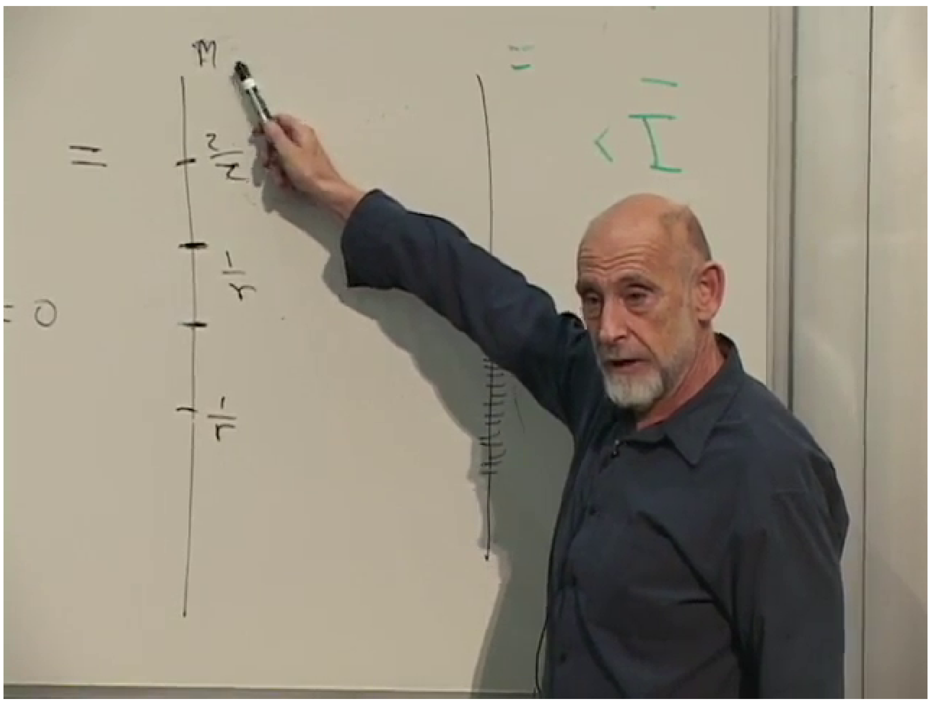

Figure 1.17: The sparse and dense mass ladders compared on the board as the radius crosses the self-dual scale.

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The exchange of the elementary mass formulas is only the visible edge of the result. Closed-string oscillator states and scattering amplitudes are also mapped consistently. A circle smaller than the string scale is not a new increasingly tiny geometry; it has an equivalent description as a larger dual circle. Point particles cannot make this identification because they possess momentum modes but no winding modes.

Question from the audience. Is this merely the same as exchanging a vertical and a horizontal direction? [a]

Response. No. It exchanges two different quantum numbers: momentum around the circle and the number of times a closed string wraps the circle. The radius must be inverted at the same time.

[a] Lecture timestamp: 01:44:38.

Question from the audience. Do the descriptions at $R < \sqrt{\alpha'}$ and $R > \sqrt{\alpha'}$ refer only to bosons, or also to fermions? [a]

Response. The duality is a symmetry of the full compactified string theory, not just the simple bosonic mass ladder used to expose it. The detailed mapping of sectors depends on the theory, but the small–large equivalence is not restricted to one particle species.

[a]Lecture timestamp: 01:46:30.

Question from the audience. Are the two states being compared open strings or closed strings? [a]

Response. They are closed-string states. A freely propagating closed string can wind a compact circle. An ordinary open string with free endpoints does not carry the same conserved winding number.

[a]Lecture timestamp: 01:47:32.

[4]

Question from the audience. Is the interchange analogous to the exchange of space and time near a black-hole horizon? [a]

Response. No. Horizon coordinates and causal character are a different issue. Here the two descriptions concern one compact spatial circle and the exchange of its momentum and winding sectors.

[a]Lecture timestamp: 01:48:00.

Question from the audience. Does this cure the short-distance self-energy problem by giving a minimum length? [a]

Response. It shows that radii below the string scale do not define new inequivalent circle backgrounds: decreasing R beyond the self-dual point increases the dual radius. This is a precise stringy obstruction to probing arbitrarily small circle geometry, although it should not be turned into a claim that every conceivable notion of distance has one universal hard lower bound.

[a]Lecture timestamp: 01:48:35.

1.11 Why Winding Sectors Cannot Be Omitted

Question from the audience. Could we define the theory using only unwound strings and leave winding states out? [a]

Response. No. Closed-string interactions generate them. Stretch an unwound loop around the compact direction so that one segment follows the circle one way and another follows it the opposite way. The net winding is still zero. A reconnection can then split the loop into two closed strings with winding $+1$ and -1. The pair may separate, so winding states must belong to the spectrum even if the initial state had none.

[a]Lecture timestamp: 01:48:56.

[4]Editorial clarification: In the fuller open-string account, T-duality exchanges Neumann and Dirichlet boundary conditions and introduces D-branes. That is why open-string T-duality is richer than simply copying the closed-string winding formula.

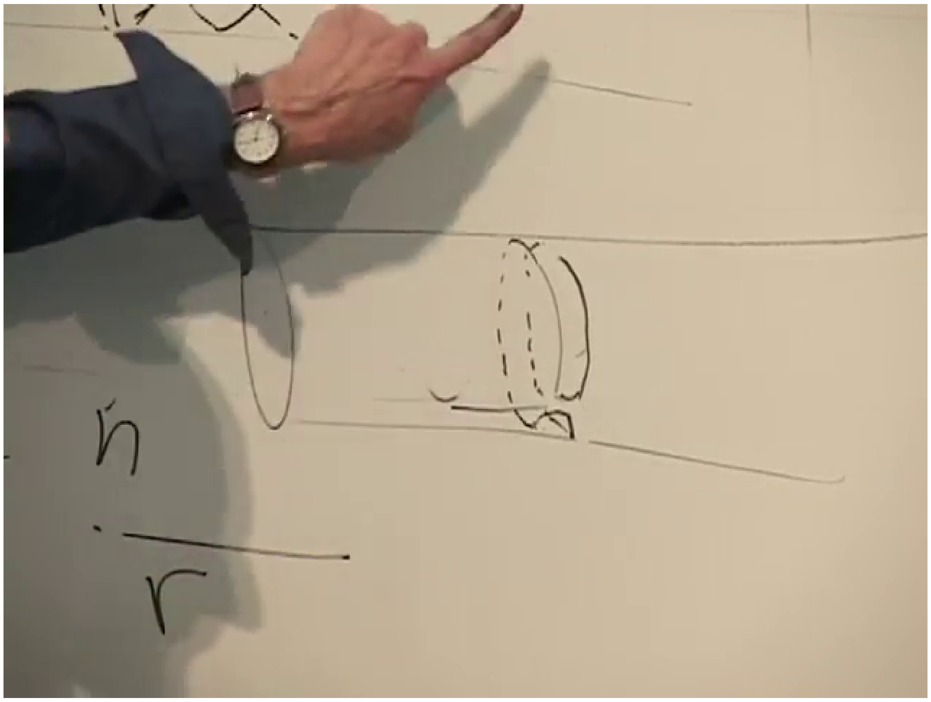

Figure 1.18: The stretched zero-winding loop arranged around the compact direction before reconnection.

Lecture frame: 01:50:30.

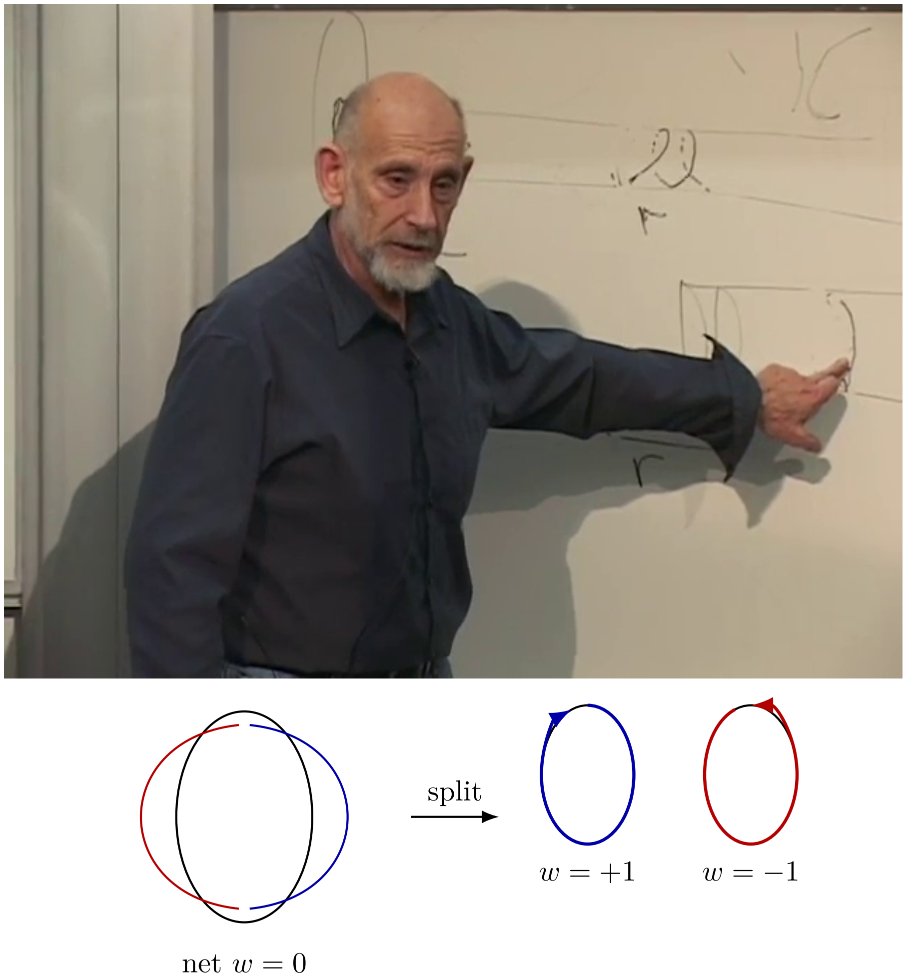

Figure 1.19: The board's separated wound loop and a clean reconstruction of winding–antiwinding pair production. Total winding remains zero.

Lecture frame: 01:51:30.

Question from the audience. Does producing the winding pair require extra energy? [a]

Response. Yes. Stretching a string costs tension energy, and the final wound strings have the masses in (1.20). A sufficiently energetic collision can produce the pair, just as a collision can produce a particle and its antiparticle while preserving the relevant conserved charge.

[a] Lecture timestamp: 01:52:11.

1.12 Can One Hear the Shape of a String Background?

For an ordinary compactified particle theory, the spacing of the Kaluza–Klein spectrum lets us infer the radius: one tries to hear the shape of the compact space from its normal modes. String theory makes the inverse problem subtler. Momentum modes hear $1/R$, winding modes hear R/α', and the complete spectrum is unchanged when the two are exchanged. Two radii that point-particle geometry regards as different are one string background described in dual variables.

The lecture has therefore arrived at both of its promised conclusions by one route. Ultraviolet consistency forces the target metric to satisfy the vacuum Einstein equation, so gravity is unavoidable. Compactification then turns hidden geometry into particle spectra, but winding states prevent that spectrum from assigning a unique classical radius below the string scale. Consistency does not merely select which geometries are allowed; through T-duality, it changes what counts as a distinct geometry at all.